

## 3D Human Avatar Quality Rating Tool

Create a dynamic 3D Human Avatar dataset with subjective quality scores using user studies.

### 1. Data acquisition

- Download public 3D human datasets (Human3.6M <sup>[1]</sup>, THuman2.1 <sup>[2]</sup>, HuGe100K <sup>[3]</sup>, etc.) covering varied subjects, body shapes, and motions.

### 2. Avatar and pose generations

- Fit or convert source data into SMPL-X <sup>[4]</sup> parameters (body, hands, face, shape  $\beta$ , pose  $\theta$ ) using NFL <sup>[5]</sup>. For hand pose estimation HaMeR <sup>[6]</sup> can also be used.
- Generate a diverse set of poses and motion sequences (animations) by sampling or interpolating pose parameters, aiming for coverage across action types, body orientations, and dynamics.
- Render the SMPL-X avatars into the target representation (mesh sequences, point clouds, or video) that will be evaluated.

### 3. Distortion and compression

- Apply motion distortions <sup>[7]</sup> (e.g., jitter, temporal downsampling or frame drops, joint noise, motion blur, smoothing artifacts) to degrade animation quality.
- Apply compression techniques <sup>[7]</sup> (mesh or point-cloud compression, video codecs, parameter quantization) at multiple bitrates or quality levels.
- Organize degradations into controlled levels so each reference has several distorted versions spanning a quality range. Document the distortion parameters for reproducibility.

### 4. Subjective quality assessment tool: Mean Opinion Score (MOS) <sup>[7]</sup> collection

- Build a rating interface where participants view the 3D avatars (with playback controls) and assign quality scores.
- Choose a methodology for quality rating.

### 5. Analysis and dataset release

- Analyze MOS distributions across distortion types and check inter-rater reliability.
- Optionally benchmark objective quality metrics against your MOS to study correlation.

## References:

[1] <http://vision.imar.ro/human3.6m/description.php>

[2] <https://github.com/ytrock/THuman2.0-Dataset>

[3] <https://github.com/yiyuzhuang/IDOL#dataset-demo-access>

[4] <https://smpl-x.is.tue.mpg.de/>

[5] <http://github.com/isarandi/nlf>

[6] <https://github.com/geopavlakos/hamer>

[7] Z. Zhang et al., "DDH-QA: A Dynamic Digital Humans Quality Assessment Database," 2023 IEEE International Conference on Multimedia and Expo (ICME), Brisbane, Australia, 2023, pp. 2519-2524, doi: 10.1109/ICME55011.2023.00429. <https://github.com/zzc-1998/DDH-QA>